

7.4 Classwork: Solving Exponential Equations (calculator)

1. [Maximum mark: 8]

Solve each equation for x . Give your answer correct to three significant figures.

(a) $3^x = 17$ [2]

(b) $4^{x-2} = 30$ [2]

(c) $2^{3x+1} = 100$ [2]

(d) $5^x = 7^{x-1}$ [2]

2. [Maximum mark: 7]

The number of bacteria P in a culture, t hours after the start of the experiment, is modelled by

$$P(t) = 200 \cdot 3^{t/10}.$$

(a) Write down the number of bacteria at the start of the experiment. [1]

(b) Find the number of bacteria after 24 hours. Give your answer to the nearest integer. [2]

(c) Find the time, in hours, at which the number of bacteria first reaches 5000. Give your answer correct to one decimal place. [4]

3. [Maximum mark: 7]

Iodine-131 is a radioactive isotope used in medicine. The mass remaining, in grams, t days after a sample is prepared, is modelled by

$$M(t) = M_0 \cdot \left(\frac{1}{2}\right)^{t/T},$$

where M_0 is the initial mass and T is the half-life. For iodine-131, $T = 8$ days. A sample initially has $M_0 = 50$ grams.

(a) Find the mass remaining after 20 days. Give your answer correct to three significant figures. [2]

(b) Find the time at which 60% of the original mass has decayed. Give your answer correct to three significant figures. [3]

(c) Find, to the nearest day, the time after which only 1 gram of iodine-131 remains. [2]

4. [Maximum mark: 6]

A cup of tea is poured. Its temperature T , in degrees Celsius, t minutes after pouring, is modelled by

$$T(t) = 20 + 65 \cdot (0.9)^t.$$

(a) State the temperature of the tea at the moment it is poured. [1]

(b) Find the temperature of the tea after 15 minutes. Give your answer correct to three significant figures. [2]

(c) The tea is comfortable to drink when its temperature reaches 40°C . Find, correct to three significant figures, the time at which this occurs. [3]